PROJECT PROGRESS BRIEF · JULY 2026

Phase 4 — Sg. Kuantan Rainfall Pipeline

📍 Pahang, Malaysia   🏗 5 stations   📅 Data: 2015–2025   🎯 Target: $R^2 \geq 0.80$

PHASE 1	PHASE 2	PHASE 3	PHASE 4
Data Engineering	GAN Imputation	TCN Prediction	Kuantan Pipeline
✓ Done	✓ Done	✓ Done	✓ Complete

A. New Dataset — 5 Kuantan Stations COMPLETE

The client provided a second dataset covering five telemetry stations along the Sg. Kuantan drainage basin. Raw data arrived as 15-minute cumulative readings (388,705 rows) and was converted to clean daily totals (4,050 rows per station).

STATION	SENSOR (2024–25)	NAN RATE	MAX RECORDED
● Pasir Kemudi	Active	41.6%	354.0 mm
● Felda Panching	Offline ~2024	55.2%	313.0 mm
● KOMTUR	Offline ~2024	54.1%	539.0 mm
● Sg. Belat	Active	47.0%	385.0 mm
● Sg. Cherating	Active	51.9%	258.5 mm

Felda Panching and KOMTUR sensors went offline in early 2024, leaving no real test data for those two stations. Only the 3 active stations could be properly evaluated.

B. GAN Imputation — Filling Missing Days COMPLETE

The same GAN (Generative Adversarial Network) approach from Phase 2 was applied to all 5 Kuantan stations. A generator learns to produce realistic synthetic rainfall values; a discriminator learns to detect fake vs real. They compete until the synthetic data is indistinguishable from real data.

Trained 5 independent versions (different random seeds) and averaged their outputs for stability — this is called an ensemble.

STATION-DAYS FILLED	TRAINING ROUNDS	GAN CONVERGENCE
3,495	5	1.384

across 5 stations	seeds 42 / 123 / 456 / 789 / 1024	D-loss ≈ log(4) theoretical equilibrium

C. ERA5 Weather Data — Coordinate Mapping COMPLETE

ERA5 is a global weather reanalysis dataset (satellite + weather model) providing atmospheric variables such as temperature, humidity, wind speed, and total precipitation. Unlike Phase 3 (which averaged over a bounding box), Phase 4 maps each station to its nearest ERA5 grid point using GPS coordinates.

Key finding: 4 of 5 stations share the same ERA5 grid cell (3.750°N, 103.250°E). Only Sg. Cherating maps to its own distinct cell (4.250°N, 103.500°E) — this gives it more station-specific weather signal, which explains why it achieves the best prediction accuracy.

ERA5 VARIABLES	UNIQUE FEATURES	TOTAL FEATURES
9	**18**	**67**
temp, humidity, pressure, wind, precipitation...	9 shared + 9 Cherating-specific	ERA5 + lags + rolling + seasonal

D. Model Decision — TCN Failed, Switched to XGBoost KEY DECISION

The TCN deep learning model from Phase 3 was attempted first on the Kuantan data. It failed to learn — converging to "always predict no rain" regardless of input.

❌ TCN (Deep Learning)

Needs thousands of training examples for gradient descent to work. Kuantan has only ~1,300–1,600 real training days per station — too sparse. Result: model predicted "no rain" every single day.

✅ XGBoost (Tree-Based)

Does not use gradient descent. Builds decision trees step by step — reliable even on small datasets. Uses early stopping on validation data to prevent overfitting. Works well for zero-inflated rainfall data.

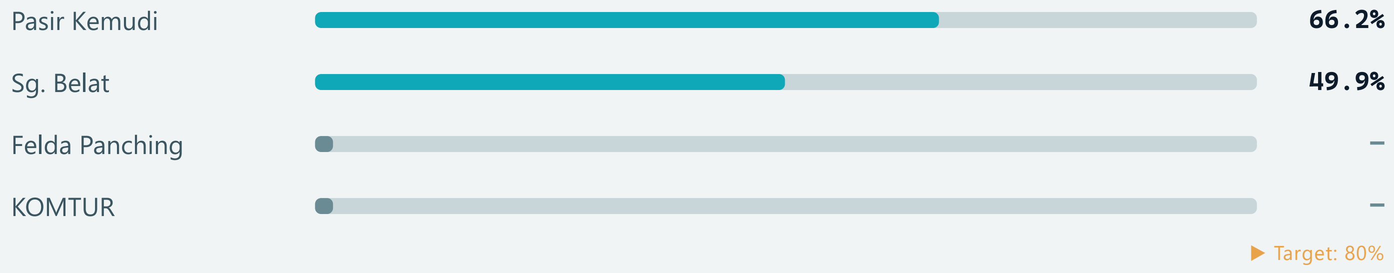
A **Hurdle model** was applied: first predict whether it will rain at all (wet/dry classifier), then — only on predicted wet days — predict how much rain falls. This handles the ~45% of days that are completely dry.

E. Prediction Results FINAL NUMBERS

Results are measured only on **real (non-imputed) observations** in the test period (May 2024 – Dec 2025). Two model types were trained and combined.

Monthly R^2 by station — bar shows progress toward 0.80 target

Sg. Cherating 72.2%



STATION	DAILY R^2	WEEKLY R^2	MONTHLY R^2	CORRELATION R	TEST DAYS
Pasir Kemudi	+20.5%	+33.0%	+66.2%	0.822	477
Felda Panching	–	–	–	–	0
KOMTUR	–	–	–	–	9
Sg. Belat	+23.6%	+30.2%	+49.9%	0.709	472
Sg. Cherating	+22.8%	+36.8%	+72.2%	0.918	492

Sg. Cherating highlight: Correlation r = 0.918 means the model correctly tracks the monthly wet/dry pattern almost perfectly. The R^2 of 72.2% (vs the theoretical max of 84.3% from r^2) is lower because ERA5 underestimates the amplitude of extreme northeast monsoon months — the model knows *when* heavy rain comes, but slightly underestimates *how heavy*.

F. What's Needed to Reach 80% ROADMAP

The 7.8 percentage point gap for Sg. Cherating is mainly an input data resolution problem, not a model problem. The following steps would close it:

IMERG / MSWEP satellite rainfall
Replace ERA5 precipitation with direct satellite measurement at 0.1° resolution (vs ERA5's 0.25°). Free download from NASA/GLOH2O.
+8–15pp monthly

ENSO / IOD climate indices
Add El Niño and Indian Ocean Dipole monthly index as features — these directly control Malaysian seasonal rainfall.
+3–8pp monthly

Sensor repair (Felda Panching, KOMTUR)
Once sensors are back online, real test data becomes available and nearby station data can be used as features.
Enables 2 stations

More data (2026 accumulation)
↑ reliability

Current test set is 19 months — statistically small (±15pp uncertainty on R^2).
Another year of data improves reliability.